Electronic Safe System

Dovydas Čepas

1 Project Description

The goal of this project was to develop an electronic safe system using Arduino Uno. The system allows the user to enter a four-digit PIN code using a potentiometer and a push button. The potentiometer selects a digit from 0 to 9, while the button confirms the selected digit.

The project has two versions: a physical Arduino implementation and a Tinkercad simulation. Both versions use the same PIN verification, locking, timing, and EEPROM functionality. The main difference is how information is displayed.

The physical version uses a MAX7219 8x8 LED matrix to display the selected digit and symbols representing different system states. In the Tinkercad version, the matrix is replaced by Serial Monitor output, which displays selected and confirmed digits as well as system status messages.

After entering four digits, the system compares the entered PIN with the PIN stored in EEPROM. If the PIN is correct, a servo motor rotates from 0 to 90 degrees to unlock the safe. The safe automatically locks after 10 seconds.

If the PIN is incorrect, an error is indicated. After three incorrect attempts, the system is blocked for 30 seconds. The PIN can also be changed by holding the button for 3 seconds while the safe is unlocked. The new PIN is saved to EEPROM and remains stored after power is disconnected.

Seven LEDs are controlled using a 74HC595 shift register. Four LEDs indicate PIN entry progress, while the remaining three indicate unlocked, incorrect PIN, and blocked states.

Parts used:

- Arduino Uno
- Potentiometer
- Push button
- Servo motor
- MAX7219 8x8 LED matrix (physical version)
- SN74HC595N shift register
- 7 LEDs and current-limiting resistors
- Breadboard and jumper wires

2 Design and Timing Budget

The system uses a non-blocking, state-based design without `delay()` calls. Five system states are implemented: PIN entry, unlocked, incorrect PIN, blocked, and PIN change.

Timer2 is configured in CTC mode with a prescaler of 64 and generates an interrupt every 1 ms. The interrupt increments a counter used to manage timing in the main loop.

The potentiometer is sampled every 100 ms. Button presses are processed using a 25 ms debounce interval. An external interrupt on pin D2 detects button presses, while the main loop handles digit confirmation and long presses.

The safe remains unlocked for 10 seconds. An incorrect PIN is indicated for 1.5 seconds, and three incorrect attempts block the system for 30 seconds. Holding the button for 3 seconds while unlocked activates PIN change mode. During PIN changes, each confirmed digit resets the 10-second timeout.

The 74HC595 shift register is controlled using direct memory-mapped I/O (MMIO) through the `PORTB` and `PORTD` registers. The button state is read directly from the `PIND` register without using `digitalRead()` or `digitalWrite()`.

EEPROM stores the four-digit PIN and a validity marker. The `EEPROM.update()` function avoids unnecessary writes to unchanged bytes.

Wiring: The potentiometer is connected to A0, the button to D2, and the servo motor to D9. The 74HC595 shift register uses D4 (DATA), D7 (CLOCK), and D8 (LATCH). In the physical version, the MAX7219 matrix uses D3 (DIN), D6 (CLK), and D5 (CS). All components share a common ground.

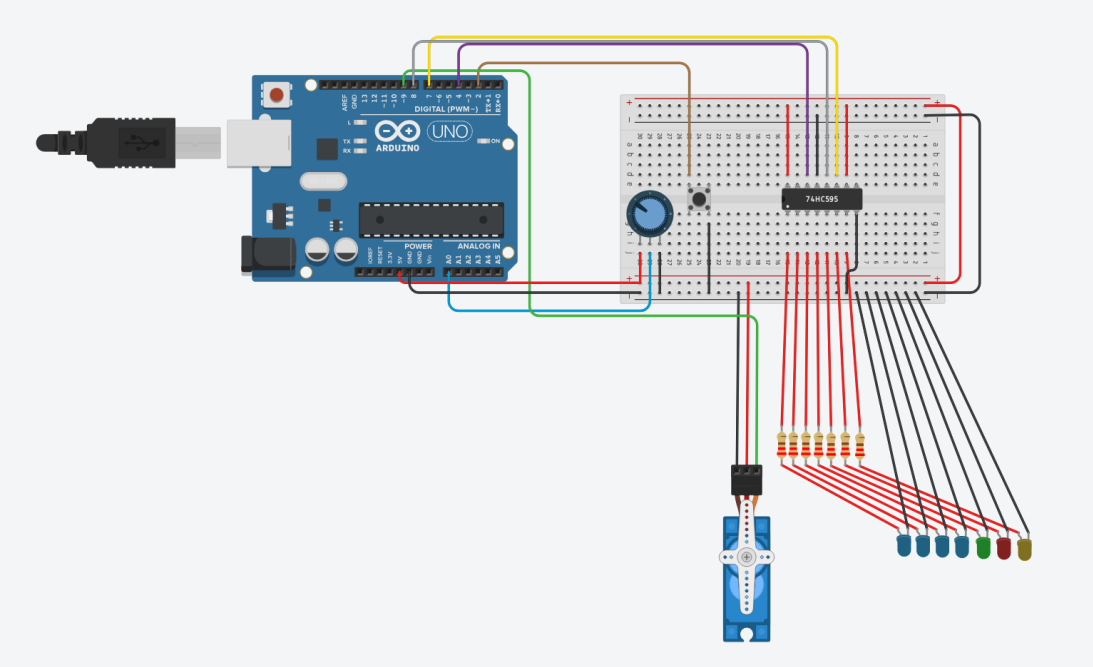


Figure 1: Physical Arduino implementation of the Electronic Safe System.

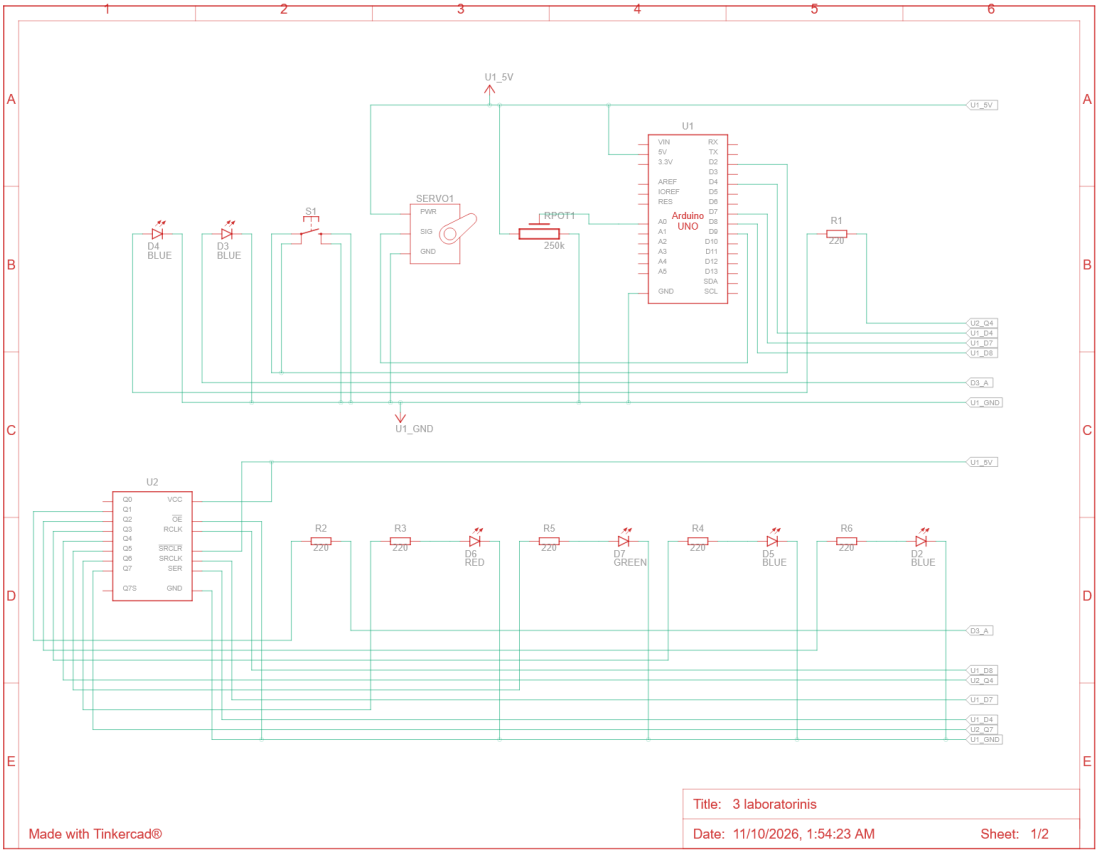


Figure 2: Tinkercad circuit layout of the Electronic Safe System.

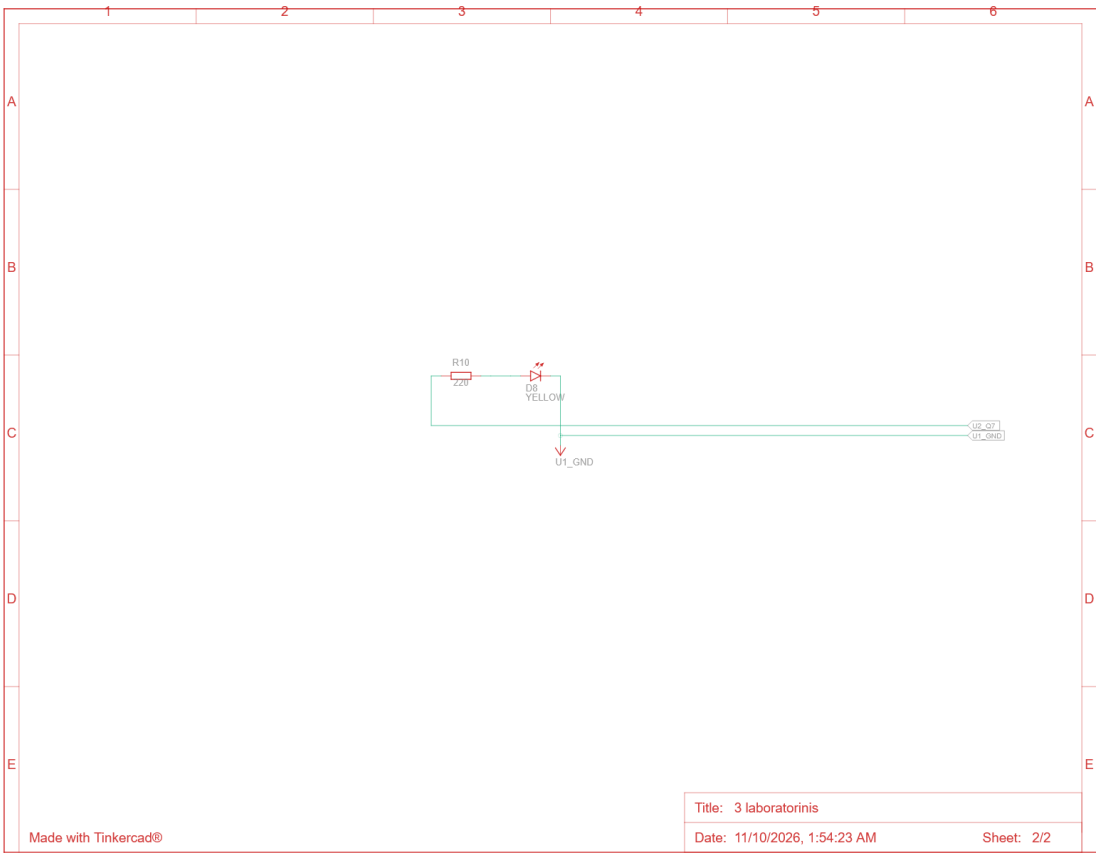


Figure 3: Tinkercad circuit schematic of the Electronic Safe System.

3 Test Results and Future Improvements

The project was implemented in two versions: a physical Arduino Uno circuit and a Tinkercad simulation. Both versions follow the same system logic and use the same PIN entry, verification, and locking mechanisms.

In the physical version, the MAX7219 LED matrix displays digits from 0 to 9 and symbols for successful PIN entry, incorrect PIN, and system blocking. Seven additional LEDs indicate PIN entry progress and the current system state.

In the Tinkercad version, the Serial Monitor replaces the LED matrix. It displays the selected digit, confirms each entered digit, and shows messages such as LOCKED, UNLOCKED, WRONG PIN, BLOCKED, and CHANGE PIN.

The system includes automatic locking, temporary blocking after three incorrect attempts, and PIN storage in EEPROM. The non-blocking design allows the system to process user input and update indicators while tracking time.

Possible future improvements include replacing the potentiometer with a numeric keypad for more convenient PIN entry and adding a physical enclosure with a locking mechanism connected to the servo motor.